

Exp no :9 Using Process Explorer to Identify Suspicious Processes

Date :

Description

- Process Explorer is a part of the Microsoft Sysinternals Suite. It is a powerful tool used to view detailed information about system processes. It helps investigators and administrators analyze active processes, detect suspicious behavior, monitor CPU and memory usage, and verify process authenticity using digital signatures.

Step-by-Step Process

Step 1: Download and Setup Process Explorer

- Go to the official Microsoft Sysinternals website: <https://learn.microsoft.com/en-us/sysinternals/downloads/process-explorer>
- Click Download Process Explorer.
- Extract the downloaded ZIP file to a folder.
- Right-click procexp64.exe (for 64-bit) or procexp.exe (for 32-bit) → select Run as Administrator.

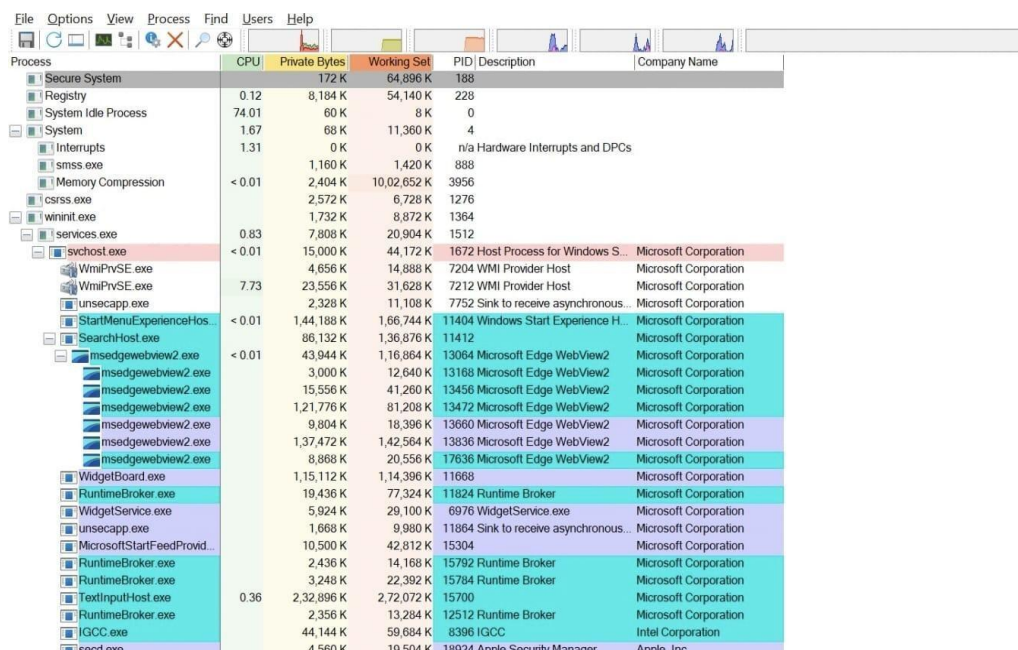
Step 2: Understand the Interface

- The main window displays all running processes in a hierarchical tree view.
- Each process shows details such as PID, CPU usage, memory usage, and company name.
- Color codes represent process states:
 - Green — Newly started processes
 - Red — Terminated processes
 - Light Blue — Processes running under the current user
 - Pink — Suspended processes

Process Name	CPU	Private	Working Set	PID	Description	Company Name
Secure System		172 K	64,896 K	188		
Registry	0.12	8,184 K	54,140 K	228		
System Idle Process	74.01	60 K	8 K	0		
System	1.67	68 K	11,360 K	4		
smss.exe	1.31	0 K	0 K	n/a	Hardware Interrupts and DPCs	
Memory Compression		1,160 K	1,420 K	888		
csrss.exe	< 0.01	2,404 K	10,02,652 K	3956		
wininit.exe		2,572 K	6,728 K	1276		
services.exe		1,732 K	8,872 K	1364		
svchost.exe	0.83	7,808 K	20,904 K	1512		
WmiPrvSE.exe	< 0.01	15,000 K	44,172 K	1672	Host Process for Windows S...	Microsoft Corporation
WmiPrvSE.exe		4,656 K	14,888 K	7204	WMI Provider Host	Microsoft Corporation
unsecapp.exe	7.73	23,556 K	31,628 K	7212	WMI Provider Host	Microsoft Corporation
SearchHost.exe		2,328 K	11,108 K	7752	Sink to receive asynchronous...	Microsoft Corporation
SearchIndexer.exe	< 0.01	1,44,188 K	1,66,744 K	11404	Windows Search Experience H...	Microsoft Corporation
msedge.exe		86,132 K	1,36,876 K	11412		
msedge.exe	< 0.01	43,944 K	1,16,864 K	13084	Microsoft Edge WebView2	Microsoft Corporation
msedge.exe		3,000 K	12,640 K	13168	Microsoft Edge WebView2	Microsoft Corporation
msedge.exe		15,556 K	41,260 K	13456	Microsoft Edge WebView2	Microsoft Corporation
msedge.exe		1,21,776 K	81,208 K	13472	Microsoft Edge WebView2	Microsoft Corporation
msedge.exe		9,804 K	18,396 K	13660	Microsoft Edge WebView2	Microsoft Corporation
msedge.exe		1,37,472 K	1,42,564 K	13836	Microsoft Edge WebView2	Microsoft Corporation
msedge.exe		8,868 K	20,556 K	17636	Microsoft Edge WebView2	Microsoft Corporation
WidgetBoard.exe		1,15,112 K	1,14,396 K	11668		
RuntimeBroker.exe		19,436 K	77,324 K	11824	Runtime Broker	Microsoft Corporation
WidgetService.exe		5,924 K	29,100 K	6976	WidgetService.exe	Microsoft Corporation
unsecapp.exe		1,668 K	9,980 K	11864	Sink to receive asynchronous...	Microsoft Corporation
MicrosoftStartFeedProvid...		10,500 K	42,812 K	15304		
RuntimeBroker.exe		2,436 K	14,168 K	15792	Runtime Broker	Microsoft Corporation
RuntimeBroker.exe		3,248 K	22,392 K	15784	Runtime Broker	Microsoft Corporation
TextInputHost.exe	0.36	2,32,896 K	2,72,072 K	15700		
RuntimeBroker.exe		2,356 K	13,284 K	12512	Runtime Broker	Microsoft Corporation
IGCC.exe		44,144 K	59,684 K	8396	IGCC	Intel Corporation
secd.exe		4,560 K	19,504 K	18924	Apple Security Manager	Apple, Inc.
APSDaemon.exe		6,116 K	25,056 K	17304	Apple Push	Apple Inc.
ApplePhotoStreams.exe		8,932 K	41,740 K	21376	iCloud Photo Stream	Apple Inc.
ShellExperienceHost.exe	Susp...	62,008 K	87,020 K	22396	Windows Shell Experience H...	Microsoft Corporation
RuntimeBroker.exe		8,116 K	42,460 K	20596	Runtime Broker	Microsoft Corporation
RuntimeBroker.exe		2,456 K	14,232 K	2196	Runtime Broker	Microsoft Corporation

Step 3: Identify Suspicious Processes

1. Look for unfamiliar or oddly named processes (e.g., xkdjeo.exe, randomname123.exe).
2. Check the Company Name and Description:
 - Legitimate software usually shows known publishers like Microsoft, Intel, or Adobe.
3. Right-click the process → Properties → go to the Image tab.
4. Verify the Path of the executable file:
 - Safe: C:\Windows\System32\ ○ Suspicious: C:\Users\<User>\AppData\Temp\ or Downloads\
5. Check for Digital Signature:
 - Valid signature = trusted developer ○ No signature or invalid = possibly malicious



Process	CPU	Private Bytes	Working Set	PID	Description	Company Name
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System	1.67	68 K	11,360 K	4		
Interrupts	1.31	0 K	0 K	n/a	Hardware Interrupts and DPCs	
smss.exe		1,160 K	1,420 K	888		
Memory Compression	< 0.01	2,404 K	10,02,652 K	3956		
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unsecapp.exe		2,328 K	11,108 K	7752	Sink to receive asynchronous...	Microsoft Corporation
StartMenuExperienceHos...	< 0.01	1,44,188 K	1,66,744 K	11404	Windows Start Experience H...	Microsoft Corporation
SearchHost.exe		86,132 K	1,36,876 K	11412		Microsoft Corporation
msedgewebview2.exe	< 0.01	43,944 K	1,16,864 K	13064	Microsoft Edge WebView2	Microsoft Corporation
msedgewebview2.exe		3,000 K	12,640 K	13168	Microsoft Edge WebView2	Microsoft Corporation
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secd.exe		4,560 K	19,504 K	18924	Apple Security Manager	Apple, Inc.

Step 4: Analyze Process Behavior

- Observe CPU, Memory, and I/O usage columns.
- If a small or unknown process consumes excessive CPU or memory, it may be malicious.
- Right-click the process → Properties → go to the TCP/IP tab.
- Check if it communicates with unknown external IP addresses.
- Examine Handles and DLLs tabs for suspicious loaded files or libraries.

explorer.exe:5748 Properties

Image Performance Performance Graph Disk and Network GPU Graph Threads TCP/IP Security Environment Job Strings

144				
TID	CPU	Cycles Delta	Suspend Count	Start Address
10724	< 0.01	26,63,233		shcore.dll!SHCreateThrea...
10660				windows.immerseshell.s...
12148				ntdll.dll!TpCallbackMayRu...
12372				Windows.UI.Xaml.dll!GetE...
8604				shcore.dll!SHCreateThrea...
11768				ntdll.dll!TpCallbackMayRu...
21456				ntdll.dll!TpCallbackMayRu...
1804				shcore.dll!SHSetValueW+
9012				CoreMessaging.dll!CoreUI...
12988				wlanapi.dll!WlanPrivateCa...
13104				wlanapi.dll!WlanPrivateCa...
2472				shcore.dll!SHCreateThrea...
13052				wlanapi.dll!WlanPrivateCa...
21940				ntdll.dll!TpCallbackMayRu...
10888				ucrtbase.dll!wcschr+0x120
3204				shcore.dll!SHSetValueW+
16816				shcore.dll!SHSetValueW+
7044				shcore.dll!SHSetValueW+
12324				shcore.dll!SHCreateThrea...
12708				shcore.dll!SHCreateThrea...
12308				ntdll.dll!TpCallbackMayRu...
12312				ntdll.dll!TpCallbackMayRu...
12300				ntdll.dll!TpCallbackMayRu...
...				...

9976				
19:49:20	21-09-2026			
Wait:WtUserRequest		8		
0:00:07.812		12		
0:00:12.812		Normal		
95,533		5		
52,60,80,17,360		2		

Step 5: Verify Process Legitimacy

1. Search the process name on Google.
- Example: svchost.exe vs svhost.exe (one letter missing — suspicious).
2. Visit <https://www.virustotal.com> o Upload the process file or search its name to verify if it's reported as malware.
 3. Cross-check with ProcessLibrary.com or official vendor websites for authenticity.



Analysing (57.2s) ...

607fc0cc3787f53af7949bb93ccefbae0157aa218b2d43fe36cb1bc95eb8d0d

Size
47.06 MB

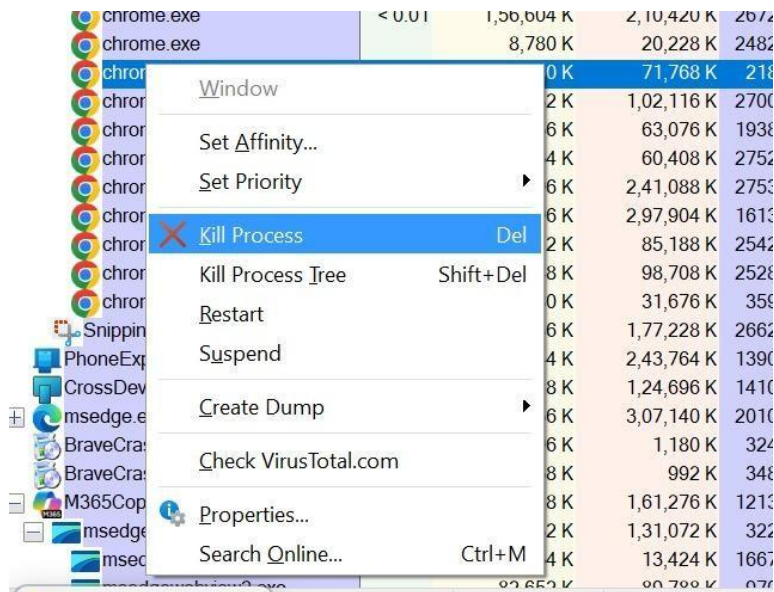
DETECTION

Join our Community and enjoy additional community insights and crowdsourced detections, plus an API key to automate checks.

Security vendors' analysis			
Acronis (Static ML)	✓ Undetected	AhnLab-V3	✓ Undetected
AliCloud	✓ Undetected	ALYac	✓ Undetected
Antiy-AVL	✓ Undetected	Arcabit	✓ Undetected
Arctic Wolf	✓ Undetected	Avast	✓ Undetected
Avira (no cloud)	✓ Undetected	BitDefender	✓ Undetected

Step 6: Take Appropriate Action

1. If the process is confirmed malicious:
 - o Right-click the process → Kill Process to stop it.
 - o Delete the corresponding executable file from its path.
2. If unsure: o Right-click → Suspend Process to stop it temporarily for investigation.
3. After removal:
 - o Run a Full System Scan using Windows Defender or Malwarebytes to ensure no remnants remain.



Step 7: Example Observation

You find faangpath_simple_template.pdf consuming 70% CPU.

- Path: C:\Users\Admin\AppData\Temp\faangpath_simple_template.pdf
- Digital Signature: None
- Company Name: Unknown
- Network Activity: Shows connections to unknown IPs in the TCP/IP tab
- Online Check: VirusTotal confirms it as a known trojan
- Action Taken: Suspended → Killed → Deleted file → Performed full antivirus scan



Rubrics:

Criteria & Marks Assigned	Mark Allotted	Mark Awarded
1. GitHub Activity & Submission Regularity	3	
2. Application of Forensic Tools & Practical Execution	3	
3. Documentation & Reporting	2	
4. Engagement, Problem-Solving & Team Collaboration	2	
Total	10	

Result

Using Process Explorer, suspicious processes were successfully identified by examining their CPU usage, path, digital signature, and network activity.

Confirmed malicious processes were terminated and removed to maintain system integrity.